3) Boiling and melting points of hydrogen is very less than that of the elements of group IV-A.

Resemblance Of Hydrogen With Halogens

1) Both Hydrogen and Halogens require one electron to complete their valence shell.
2) Both hydrogen and Halogens are non-metals(Except iodine which has partial metallic character).
3)Both hydrogen and Halogens acts as negative ions when combined with metals.

4) Their molecules are diatomic.
5) Like Halogens, hydrogen can also form both ionic and covalent bond.

Explanation

Hydrogen

NaH (ionic),
CH_4 (Covalent)

Halogens:

NaCl (ionic), HCl
6) Both H and Halogens form compound with metals and non-metals.

Explanation

Hydrogen :H_2

S [S=non-metal]
Halogens : HCl [H = non-metal]

DIFFERENCES

1) Hydrogen has one electron in valence shell while Halogens has seven electrons.
2) Electronic configuration of H = 1S1,
Valence shell electronic configuration of Halogens = nS2, nS5

3) Unlike X- ion, H- ion is unstable in water and reacts in the following manner.
Explanation**H- + H2OèH2+ OH-**

UNIQUE ATOMIC STRUCTURE OF HYDROGEN

:

Hydrogen is the only element which has no neutron in its nucleus.

Conclusion

Due to above reasons it is clear that hydrogen cannot be placed in any one of the above mentioned groups.

Position of hydrogen in periodic table

Position of an element in periodic table depends upon its electronic configuration and properties. Hydrogen resembles the elements of group I-A, IV-A and VII-A in some respects. Properties of hydrogen do not completely match any one of the above-mentioned groups. That is why position of hydrogen is still undecided.
In the coming lines we will discuss why hydrogen can not be placed in a particular group in the periodic table.***Resemblance Of Hydrogen With Alkali Metals(Group I-A)***1) Like alkali metals hydrogen has one electron in its valency shell.***Explanation*****H (Z=1): K1Li (Z=3): K2, L1Na (Z=11): K2, L8, M1**(2) Valance shell electronic configuration of hydrogen and alkali metal is same.***Explanation*****H (Z=1): 1S1Na (Z=11): 1S2,2S2,2P6,3S1**3) Both hydrogen an alkali metals are good reducing agents.4) Like alkali metals hydrogen can also form halides.***Explanation:****Hydrogen:*
HCl, HI.*I-A:*
NaCl, KBr.5) Halides of alkali metals and hydrogen ionized in similar way in aqueous solution.***Explanation***HCl (aq) èH+(aq) + Cl-(aq) NaCl(aq)èNa+(aq) + Cl-(aq)

6) Like alkali metals hydrogen can also form compounds with non-metals.***Explanation:I-A:***Na2S*Hydrogen:*H2S[S and Cl are non-metals]***DIFFERENCES :-***1) Hydrogen is a non-metal where as elements of I-A are metals.
2) Hydrogen can form both covalent and ionic compounds. But alkali metals only form ionic compounds.
3) Hydrogen cannot lose its valence electron.
4) Hydrogen is a gas at room temperature but alkali metals are solids.
5) Unlike Na+ or K+ hydrogen ion (H+) is unstable in water.*Explanation:***H+ + H2OèH3O+**6) I.P. of hydrogen is very high as compared to alkali metals.***Explanation***

I.P. of hydrogen is 313 Kcal per mole.
I.P. of alkali metals max. is 147 Kcal per mole.

Resemblance of Hydrogen with Carbon Family (IV-A)

1) Valance shell of hydrogen is half-filled like the elements of group IV-A.***Explanation:***

H (Z=1) : Capacity of K-Shell =2 , no. of electrons = 1 , % = 50%

C(z=6) : Capacity of L-Shell = 8 ,no. of electrons

= 4 , % = 50%2) I.P. and E.A. of hydrogen and carbon are comparable.3) E.N. of H and Carbon family are almost similar. [

H = 2.1, C = 2.5 , Si= 2.4 ***Explanation***

DIFFERENCES1) Carbon is tetravalent but Hydrogen is a mono-valent.

2) Hydrogen is a gas at room temperature but elements of group IV-A are solids.